When concord meets suppletion: on double plural marked adjectives in Ingush

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Abstract This paper examines the patterns of apparent double plural marking on the Ingush adjective *-oaqqa/-oaqqii* 'big'. It presents evidence against Norris's (2022) analysis of the non-prefixal plural marking on this adjective as concord. New data demonstrate that (a) prefixal and non-prefixal plurality may mismatch, and (b) such mismatches are disallowed in certain idiomatic contexts. The alternation *-oaqqa/-oaqqii* is treated as an instance of number suppletion and is analyzed as word-external contextual allomorphy. The paper discusses whether – and to what extent – the Ingush data in question can be seen as evidence against analyzing adjectival number suppletion as a concord-mediated operation.

1. Introduction

One interesting problem in the study of the morphology of adnominal modification concerns the cases where the plurality is exponed more than once on the adnominal modifier (Deal (2016), Norris (2022)). One particular case of such a position was discussed recently in Norris (2022) and concerns the morphologically irregular adjective *-oaqqa/-oaqqii* 'big' in Ingush[1]. In particular, in plural contexts, this adjective may have the form *boaqqii* (or *joaqqii* or *doaqqii*, depending on the noun class), where plurality is exponed both on the prefix *b-* (cumulatively with noun class) and by means of choosing the form *-oaqqii*[2] over its non-plural allomorph *-oaqqa.* A minimal pair containing this adjective in singular and in plural context is provided below in (1). Norris (2022), which surveys theoretical approaches to nominal inflection in Distributed Morphology, analyzes this fact by treating both plural exponents on the adjective *boaqqii* as instances of concord. To that effect, the form *boaqqii* is analyzed as exhibiting concord twice: on the prefix and on the segment *-ii*. Among other things, this analysis suggests a morphosyntactic rule (the insertion of the number agreement node) that applies to a single Ingush adjective (viz. *-oaqqa* 'big') [3].

(1) v.oaqqa	sag	/	b.oaqq-ii	nax
V.big(SG)	man(SG)	/	B.big-PL	man(PL)
'big man' / 'elders'				

(Norris (2022), citing Nichols (2011))

[1] Ingush is an Ergative SOV Nakh-Dagestanian language of the Nakh group spoken predominantly in Ingushetia, Russia. It is a relatively well studied language, an interested reader can be referred to Nichols (2011) on more information on Ingush grammar.
[2] Note that the string *-oaqqii* is analyzed as bimorphemic, analyzing *-ii* as a plural suffix by Norris. This particular assumption will be challenged in the subsequent exposition (see e.g. ft. 11 on p. 9).
[3] All Ingush data in the paper, unless otherwise is indicated, comes from the author's own fieldwork. All examples have been checked with at least three native speakers.

In this paper, I present data that challenge the view that the two instances of plural marking in forms like *boaqqii* reflect the same morphological (or morphosyntactic) operation and argue instead that while prefixal plurality is indeed concord, the non-prefixal plurality is best analyzed as an instance of word-external suppletion.

More specifically, I present two novel empirical generalizations concerning the distribution of non-prefixal plural marking on Ingush adjectives. First, the prefixal and non-prefixal plural exponents on the Ingush adjective -*oaqqa/-oaqqii* may mismatch in (most) plural, but not singular contexts, as the examples in (2) show.

(2) **Adjectival plural marking in Ingush**

a. b-oaqqa q'oalam
B-big pen
'a big pen'
b. d-oaqqa/d-oaqqii q'oalam-až
D-big/D-big.PL pen-PL
'big pens'

Second, the mismatch is disallowed in several specialized contexts, namely in idiomatic collocations with the noun *sag/nax* 'person/people', where the mismatch is disallowed and the plurality must be exponed both prefixally and non-prefixally. To this effect, the following string (3a), where the non-plural form -*oaqqa* is used, is grammatical, but cannot have the intended idiomatic reading, while in (3b), where the plural form -*oaqqii* is used, both idiomatic and non-idiomatic readings are available.

(3) a. *b-oaqqa nax* 'big people'; *'elders'

b. *b-oaqqii nax* 'big people'; 'elders'

Relatedly, the distributional pattern of non-prefixal marking in the Ingush adjective *big* (that is, the distribution of -*oaqqa* vs. -*oaqqii*) is identical, in relevant aspects, to that found in the number paradigm of the fully suppletive adjective *zwamig/kegii* 'small'. This adjective, similar to -*oaqqa/-oaqqii*, allows for the free distribution of the allomorphs in most plural contexts, and requires the choice of the plural allomorph (*kegii* over *zwamig*) in idiomatic collocations with the noun *sag*/*nax* 'person/people' in the plural, as shown below in (4).

(4) a. zwamig/kegii q'oalam-až
small/small.PL pen-PL
'small pens'
b. zwamig nax

small people

'small people'; '*youngsters'

In this paper, I analyze these facts as follows. I suggest that in the example (3b), only the prefix (*b*- 'PL') is a result of concord. The non-prefixal plural marking (that is, the choice of *-oaqqii* over *-oaqqa*) is a result of allomorphic selection, which operates within the nominal domain but is sensitive to locality. To that effect, the same mechanism (allomorphic selection) is responsible for the distribution of the two number pairs of adjectives in the language: *zwamig/kegii* and *-oaqqa/-oaqqii*.

The apparent number mismatch in (3a) is thus a result of a discrepancy in the application of the two operations. While prefixal number marking is a result of concord, which is obligatory on all prefix-taking adjectives, the non-prefixal number marking is a result of allomorphic selection, which shows a different distribution (roughly, it may be optional in some cases, the details are provided below).

The rest of the paper is organized as follows. In section 2, after introducing the basics of Ingush morphosyntax, I discuss the main data, including the patterns of plural marking on the Ingush adjectives *big* and *small* in idiomatic and non-idiomatic contexts. In section 3, I suggest an analysis that treats non-prefixal number marking in the two Ingush adjectives in question as word-external allomorphy. Finally, in section 4, a theoretical question is raised of whether the Ingush data provide a piece of evidence for analyzing adjectival number suppletion as agree/concord non-mediated allomorphy. I conclude that an agree/concord-mediated analysis for Ingush adjectives can only be maintained with a set of very specific and theoretically implausible assumptions. Section 5 concludes.

2. Data: prefixal and non-prefixal plural marking in Ingush adjectives

The main claim of this section is that prefixal and non-prefixal plural marking on Ingush adjectives exhibit significantly divergent distributional properties, warranting analysis as distinct phenomena. Before detailing the relevant pattern, I first outline key aspects of Ingush morphosyntax.

In Ingush, number on adnominal modifiers is primarily marked via prefixes that cumulatively mark noun class (gender) and number. The language distinguishes two numbers (singular, plural) and six noun classes, yielding four distinct noun class/number prefixes. Every Ingush noun belongs to one of these six classes. Roughly 10% of adjectives (per Norris (2022)) host a prefixal slot; the rest lack it. As Nichols (2011) details, the presence of such prefixes on modifiers is purely morphological, with no impact on their syntactic behavior. Examples for each class appear below in (5) (Table 1 provides the full prefix inventory; note that prefixless adjectives do not exhibit prefixal inflection irrespective of the class of the noun).

(5) **Prefix- and non-prefix taking adjectives**

a. dika q'oalam

good pen

'a good pen'

b. b-weaxa q'oalam

B-long pen

‘a long pen’

(6) **Table 1: noun class prefixes in Ingush**

	Singular	**Plural**
Class 1	v-	d-/b-
Class 2	j-	d-/b-
Class 3	j-	j-
Class 4	d-	d-
Class 5	b-	b-
Class 6	b-	d-

2.1 Suppletion in adjectives: *big* and *small*

In addition to prefixal marking, two Ingush adjectives, namely *-ooaqa/-aoqqii* ‘big’ and *zwamig/kegii* ‘small’, exhibit non-prefixal number marking[4].

I start with the adjective -*oaqqa/-oaqqii* ‘big’. This adjective belongs to the class of adjectives that contain a morphological slot for a noun class/number prefix. This means that any form of this adjective must contain a prefix which cumulatively marks noun class and number of the modified noun. Apart from co-varying in noun class/number with the head, the adjective also appears in two forms: -*oaqqa* and -*oaqqii*. The form -*oaqqii* is universally found in plural contexts, such as in the examples below.

(7) **The adjective –*oaqqa/-oaqqii* with B-D class (basic paradigm; to be refined)**

a. b-oaqqa/*b-oaqqii q'oalam

B-big /B-big.PL pen

‘big pens’

b. d-oaqqii q'oalam-až

D-big.PL pen-PL

‘big pens’

In most descriptive works on Ingush morphosyntax, such as Nichols (2011) and Norris (2022), the -*oaqqa/-oaqqii* alternation is taken as a simple singular/plural pair that reflects the number of the head noun. In such a way, the -*oaqqii* form is seen as doubling – at least, from a descriptive perspective – the plural feature cross-referenced on the prefix. Such works, however, fail to observe a more intricate pattern which governs

[4] Certain individual non-adjectival modifiers can also have suffixal plural marking: *ši – ši’-až* ‘two’ — ‘two.PL’ *jer – jer-až* ‘this’ — ‘these’. The suffix -*až* is also a regular plural suffix for nouns.

the distribution of *-oaqqa* vs. *-oaqqii*. More specifically, it appears that while the noun class/number prefix always tracks the number (as well as noun class) of the head noun, both *-oaqqa* and *-oaqqii* in most plural contexts are in free variation (see (8) below). That is to say, in most cases[5] speakers accept both morphological forms in plural contexts, failing to specify any tangible difference between them. The form *-oaqqii* – on the other hand – can only be used in plural contexts and using this form in singular contexts results in ungrammaticality (8a). Importantly**, the distribution of *-oaqqa* vs. *-oaqqii* is different from the distribution of singular vs. plural feature on the prefix**.

(8) **Distribution of *-oaqqa vs. -oaqqii* (refined)**

a. b-oaqqa/*b-oaqqii q'oalam
B-big/*B-big.PL pen
'a big pen'

b. d-oaqqa/d-oaqqii q'oalam-až
D-big/D-big.PL pen-PL
'big pens'

A full morphological paradigm of this adjective is provided below. (Note that a single prefix may serve several noun classes, e.g. *b-* appears in the singular of both Class 5 and Class 6).

(9) **Table 2. Full morphological paradigm of *-oaqqa/-oaqqii* (excluding case inflection)**

	Singular	**Plural**
Prefix J-	*j-oaqqa*	*j-oaqqii*
Prefix V-	*v-oaqqa*	N/A
Prefix D-	*d-oaqqa*	*d-oaqqii*
Prefix B-	*b-oaqqa*	*b-oaqqa*

Of particular interest for the current discussion is the example in (10), where the form *-oaqqa* is used while the plural prefix is selected. Importantly, the singular marking on the prefix results in ungrammaticality. This example shows that plural marking on prefixes, which I analyze as concord, has a different distribution compared to the distribution of the *-oaqqii* allomorph.

(10) **Double plural marking on the adjective**[6]

b-oaqqa/*d-oaqqa q'oalam-až

[5] See the discussion of idiomatic contexts, as well as register differences, below.

[6] For readers' convenience, I will henceforth gloss prefixes simply as SG or PL, abstracting away from particular inflectional classes.

PL-big/*SG-big pen-PL
'big pens'

Furthermore, there exists a set of specialized plural contexts where the use of the non-plural form -*oaqqa* is not allowed and the plural form -*oaqqii* must be used instead. More specifically, the plural marking is obligatory in non-compositional expressions with the noun *sag* (pl. *nax*) 'person'[7]. The expression has the non-compositional meaning *elder*, *senior person (most commonly man)*[8]. The consultants remark that on compositional readings, the expression isn't frequently used in the language since the adjective -*oaqqa* is not naturally used when referring to a person's body size. However, in specially constructed contexts in which the word *sag* is modified by the adjective, the form -*oaqqa* is judged grammatical on compositional construals. One type of contexts deemed acceptable by all consultants involves two drawings of human beings, where one drawing is significantly larger than the other. The large drawing (i.e. the 'person' in this drawing) can be referred to as *v-oaqqa sag* 'big person'.

The following examples show that while the allomorph -*oaqqii*, when used with the word *nax* 'people', can yield both idiomatic and compositional readings, when the -*oaqqa* allomorph is chosen in a plural context, the idiomatic reading is lost and only the compositional reading is available.

(11) **Distribution of *-oaqqa vs. -oaqqii* in expressions with *sag/nax* 'person/people'**

a. *v-oaqqa sag* 'big person' or 'elder'

b. *b-oaqqii nax* 'big people' or 'elders'

c. *b-oaqqa nax* 'big people'; '*elders'

To that effect, the following generalization emerges. In idiomatic contexts choosing the allomorph -*oaqqii* is obligatory.

(12) **Generalization**

a. In non-compositional plural contexts, the plural form is obligatory
b. In compositional plural contexts, the plural form can be used interchangeably with the singular (number-neutral) form

The distribution of the two allomorphs is summarized in the following table.

(13) **Table 3**

[7] The fact that the noun *sag* has itself a suppletive number paradigm is completely accidental and has no bearing on the problem in question.
[8] See Nichols (2011:101; 221) for some discussion.

	Singular compositional	**Plural compositional**	**Singular non-compositional**	**Plural non-compositional**
-oaqqa	possible	possible	possible	impossible
-oaqqii	impossible	possible	impossible	possible

Before proceeding to the analysis of the allomorph distribution I turn to discussing number marking in another adjective with an irregular number paradigm – the adjective *zwamig/kegii* ‘small’. The two main differences, compared to the already discussed adjective *-oaqqa/-oaqqii* are the following. First, unlike the adjective *-oaqqa/-oaqqii*, the adjective *zwamig/kegii* does not contain a morphological slot for a noun class/number prefix. Second, unlike the adjective *-oaqqa/-oaqqii* the number paradigm of *zwamig/kegii* is fully suppletive, see the basic paradigm below:

(14) **Basic paradigm of *zwamig/kegii* (to be refined)**

a. zwamig q'oalam
small pen
‘a small pen’

b. kegii q'oalam-až
small pen-PL
‘small pens’

What is remarkable about the adjective *zwamig/kegii* is that its suppletive pattern closely mirrors the distribution of the *-oaqqa* vs. *-oaqqii* forms. More specifically, in most plural contexts, both suppletive forms are allowed. This is illustrated in the following example, where both forms can be used with the plural noun and only the form *zwamig* can be used in the singular context.

(15) **Paradigm of *zwamig/kegii* (refined)**

a. zwamig/*kegii q'oalam
small/small.PL pen
‘a small pen’

b. zwamig/kegii q'oalam-až
small/small.PL pen-PL
‘small pens’

However, in idiomatic expressions with the noun *sag (nax)*, where the whole expression has the meaning *youngster, young person*, the specialized plural form *kegii* is used[9].

[9] Nichols (2011: 221) notes that *kegii* is only compatible ‘with a few words’, without specifying the exact list. A brief google search has revealed at least two clear cases of nouns compatible with *kegii*, apart from *sag/nax*: *kegii beraž* `small children’ and *kegii wajgaž* `teaspoons (i.e. small spoons)’. My consultants remark that *kegii* is compatible with a very wide range of adjectives and fail to specify any obvious constraint in its modificational compatibility.

(16) **Paradigm of *zwamig-kegii***

a. *zwamig sag* ‘small person’ or ‘youngster’

b. *kegii nax* ‘small people’ or ‘youngsters’

c. *zwamig nax* ‘small people’; ‘*youngsters’

Similar to the distribution of *-oaqqa* vs. *-oaqqii*, the specialized plural form of the adjective, viz. *kegii*, is reserved for plural contexts, the form *zwamig* can be used in the plural in non-idiomatic contexts, interchangeably with the form *kegii*, and is unavailable in plural idiomatic contexts, where the only acceptable form is *kegii.*

Below I provide a table which generalizes the distribution of the number allomorphs in the two discussed adjectives.

(17) **Table 4.**

	Singular	**Plural**
compositional	*-oaqqa; zwamig* **-oaqqii; *kegii*	*-oaqqa; zwamig* *-oaqqii; kegii*
non-compositional	*-oaqqa; zwamig* **-oaqqii; *kegii*	**-oaqqa; *zwamig* *-oaqqii; kegii*

The two adjectives are the only irregular adjectives reported in the available resources on Ingush (Malsagov 1963, Nichols 2011), these are also the only ones that I could find in the course of elicitations. The rest of the adjectives either show no morphological number distinctions or exhibit prefixal noun class/number marking, similar to the one on the adjective *-oaqqa/-oaqqii*.

2.2. Double plurality: concord vs. suppletion

Let’s summarize the discussion. We have seen that while plural marking always tracks the number of the head noun, the distribution of non-prefixal number marking on the adjective *-oaqqa/-oaqqii* is different, since the choice of non-prefixal non-plural marking is allowed in plural non-idiomatic contexts, such as (8b). Furthermore, the distribution of non-prefixal plurality in *-oaqqa/-oaqqii* is identical, in all relevant aspects, to the distribution of the number-conditioned allomorphs (*zwamig/kegii*), which suggests that the forms *-oaqqa/-oaqqii* can be plausibly analyzed as a (partially) suppletive pair. If on the right track, this line of research can give a principled account of the problem raised by Norris (2022) concerning the nature of double plural marking in Ingush. To wit, the prefixal plural is an instance of concord, while the non-prefixal plural marking is an instance of allomorphic selection.[10]

In the subsequent section, I provide a formal morphosyntactic account of Ingush adnominal plural marking.

3. Analysis

3.1. Assumptions

In this analysis, I adopt the following assumptions.

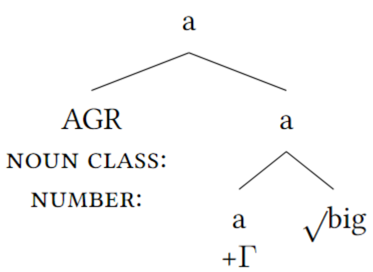
I follow Norris (2022) in treating prefixal number marking on adnominal modifiers as a result of concord and take facts like (8) as an indication that concord is obligatory in the Ingush Noun Phrase for prefix-taking modifiers. For non-prefix-taking modifiers, I suggest a lack of concord relation although one can alternatively suggest an abstract concord operation – the choice between the two options is not crucial and nothing in my analysis hinges on this choice.[11]

More specifically, I suggest that the prefix is the spellout of a node merged postsyntactically, at the PF computation, which adjoins to the complex head, which consists of the root and the categorizing head *a(djective)*, see the diagram below. In the subsequent exposition, I will concentrate on plural marking, while the mechanics of noun class inflection is supposed to proceed analogously with no remarkable detail worth separate discussion. Building on Norris (2022), I suggest that a subset of Ingush adjectives, viz. those that take a prefix, have a diacritic feature +Γ which triggers an insertion of an AGR node which has unvalued noun class and number features that get valued from a dominating source. To sum up, the described picture presents an ordinary and perhaps typologically unremarkable position where concord proceeds obligatorily at all modifiers which contain the corresponding morphological slot (i.e. the prefix).

(18) **Structure for concord (adopted from Norris 2022)**

a. AGR-insertion: $X_{[+\Gamma]} \rightarrow [X_{[+\Gamma]}\ \mathrm{AGR}]_X$

b.



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[10] Relatedly, both adjectives are size adjectives, which is a typical semantic domain for number suppleting modifiers (See e.g. Veselinova (2005)).

[11] However, the proposed analysis departs crucially from Norris (2022) in assuming that *-oaqqii* is a monomorphemic string and does not contain a plural suffix.

Syntactically, adjectives with compositional semantics merge in a distinct position within the NP from those with non-compositional semantics. The primary motivation is that compositional adjectives invariably precede non-compositional ones; see (21) below.

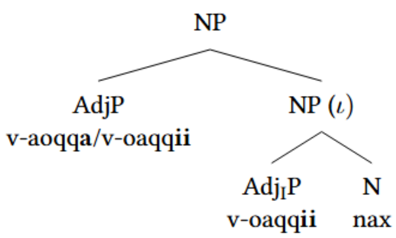
(19) **Idiomatic adjectives linearly follow non-idiomatic adjectives, non-idiomatic marked bold**

a. ***b-oaqqii*** *b-oaqqii nax* or *b-**oaqqa*** *b-oaqqii nax* '[physically] big elders'

b. ***kegii*** *b-oaqqii nax* or ***zwamig*** *b-oaqqii nax* '[physically] small elders'

For notational simplicity, I label the domain hosting non-compositional (idiomatic) adjectives as ι (iota). I remain deliberately agnostic about any relation between ι and other established NP domains (e.g., phasal ones).

(20) **The idiomatic domain (ι) within the NP**



Attributing distinct attachment heights to idiomaticity/compositionality allows us to analyze allomorph selection in Ingush adjectives as structurally local (rather than merely linearly adjacent). For instance, allomorph distribution cannot be reduced to linear adjacency to the head noun, as both idiomatic and non-idiomatic adjectives can be linearly adjacent while retaining their suppletive forms. In the example (23), the non-idiomatic adjective *can* (though need not) appear in its plural suppletive form, despite being non-adjacent to the head noun.

(21) **Linear adjacency is not required for number suppletion**

kegii/zwamig b-oaqqii nax '[physically] small elders'

The data from word order shows that compositional adjectives can either precede or follow the numerals, while non-compositional adjectives must strictly follow numerals, which might provide additional evidence for the structural difference between the two types of adjectives.[12]

Adjectives with numerals

a.	ši'	v-oaqqa	sag
	two	SG-big	person

[12] The consulted speakers report a weak preference for ADJ-NUM-N orders (which are also apparently the prescribed orders) for compositional adjectives while for non-compositional adjectives the NUM-ADJ-N order remains strict. I leave for future research investigating the nature of this variation.

‘two elders’, ‘two big persons’

b. v-oaqqa šiʼ sag

SG-big two person

‘two big people’, *‘two elders’

I assume a simple NP structure where adjectives are (at least on the surface) nominal modifiers. I remain agnostic on whether nouns are generated inside AdjPs as adjectives’ subjects (Stowell 1983, Kratzer 1995 and subs.[13]) or whether adnominal AdjPs involve relative clauses (see e.g. Alexiadou (2014)).

As a relevant detail, both types of morphological number marking (that is, prefix marking and allomorphy/suppletion) are sensitive to syntactic, rather than semantic, number. This can be evidenced, for example, by (22), where the morphologically singular, though semantically plural, noun triggers singular across the two types (viz. prefixal and allomorphical) of adnominal morphological marking. In this example, which involves numerical expressions, where morphologically singular noun must be used, both plural marking on the prefix and selecting the plural allomorph are impossible. Also, as this example shows, the singular, rather than plural, allomorph of the noun *sag/nax* must be selected.

(22) **Adnominal modification with numerical expressions**

šiʼ v-oaqqa/*b-oaqqa/*v-oaqqii/*b-oaqqii sag

two SG-big/*PL-big/*SG-big.PL/*PL-big.PL person

‘two elders’; ‘two big persons’

Morphologically, I adopt core tenets of Distributed Morphology (Halle & Marantz 1993; Bobaljik 2012; Moskal 2015; Bobaljik & Harley 2017; Adamson 2024). Following Embick (2010, 2015), Bobaljik (2012), a.o.. This approach presupposes that suppletive alternations are to be modeled as morphemic alternations which proceed postsyntactically. Different allomorphs compete for the same position according to the Subset Principle and are sensitive to their environment; the allomorphs are assumed to be sensitive to structural Bobaljik (2012); Choi and Harley (2019) rather than linear Embick (2010) adjacency. A scheme of such modeling is provided below.

(23) **A scheme of an allomorphic rule**

√ROOT ↔ b / [x]

√ROOT ↔ a

The irregular forms are taken to be context-dependent allomorphs (Bobaljik 2012, 2015; Harley 2014, 2015; Arregi and Nevins 2014, Lomashvili 2019, a.o.) of the same vocabulary item, or root (hence Same Root Analysis (Adamson 2024)). This view relies on the assumption that the allomorphy can not only be conditioned by internal inflectional features of a phrase (such as number-conditioned allomorphy in English

[13] In such a case the structures in (20) can be seen as a notational simplification of a more complex structures where the subject is contained inside the AdjP.

person-people), but also by active morphosyntactic features of other syntactic nodes in its domain (Bobaljik and Harley 2017). As for locality, I take the position that strict linear adjacency is not a prerequisite for contextual allomorphy and that allomorphy is sensitive to local syntactic domains, see the specific implementation below.

Finally, I take the similarity of the distributions of *zwamig/kegii* vs. *-oaqqa/-oaqqii* as evidence that the string *-oaqqii* must be analyzed as one morpheme (that is, alternating stem), similar to *kegii*, where *-ii* is not a separate (viz. plural) morpheme.

3.2. √SMALL

I start by analyzing the suppletion in *zwamig* vs. *kegii*. To sum up the data, we have seen that in singular environments the singular (non-plural) allomorph must be selected irrespective of whether the adjective is used compositionally or not. In plural environments, the plural allomorph must be used with idiomatic adjectives and the singular (number-neutral) allomorph can be used along with its plural counterpart in non-idiomatic plural contexts. The distribution of the two allomorphs clearly suggests that *zwamig* has a less specified environment and should be considered an elsewhere form. I will start with the rule, which I will use as a baseline and which will be refined in the subsequent exposition.

(24) **Rule for** √SMALL **(to be refined)**

√SMALL ↔ kegii / [-sg]

√SMALL ↔ zwamig

Although this rule correctly predicts that *zwamig* will be used in singular environments, it crucially does not predict the sensitivity of the allomorphs to idiomaticity nor the free variation of the two allomorphs. In order to model this, I suggest that a locality-sensitive rule exists in Ingush, although locality may be computed in two different ways. More specifically, I suggest that the locality is computed with respect to the closest locality boundary which may be optionally deleted. Most directly, this concerns the boundary of the ι domain, which, to recall, comprises the part of the structure, where adjectives with non-compositional meanings are merged. If deletion does not proceed, the compositional adjectives end up left outside of the domain with the allomorph-selecting rule not applying to them. On the contrary, if the boundary remains deleted by the end of the derivation, the high (i.e. compositional) adjectives end up in the same domain as non-compositional adjectives; which results in both types of adjectives possessing the same degree of allomorphic sensitivity to the number features of the head. This, in its turn, forces such adjectives to select the plural rather than the number-unmarked allomorph with the plural noun.

The rule suggested below suggests that the relevant locality domain, to which the allomorphic rule is sensitive, may constitute either the ι (in the case where the domain boundary is not deleted), or the upper NP(DP) boundary – in the case where the ι boundary is deleted.

(25) **Allomorph Selection Rule √SMALL (ω – locality domain)**

√SMALL ↔ kegii / [_ N[-sg]]$_{\omega}$

√SMALL ↔ zwamig

The rule in (26) selects for *kegii* only if the head noun is plural and if the environment and the context are contained within the same domain. If the ι boundary is not deleted, then for the contexts including the plural compositional readings, the elsewhere form is selected. If the ι boundary is deleted, a more symmetrical allomorphic space is created where the choice between the two allomorphs depends solely on the number of the head noun.

The optionality of *zwamig* vs. *kegii* arises, in this system, due to the fact that in the language, the domain boundary can be optionally deleted. For a subset of environments, that is, for plural non-compositional contexts, the deletion of the boundary has zero consequences for the allomorph selection since, for instance, both rules yield *zwamig* in singular environments and both rules yield *kegii* in plural idiomatic contexts. In contrast, for a subset of environments, viz. for plural compositional contexts (i.e. the higher adjectives), this choice affects the allomorph selection.

As I have already stated above, I have failed to find clear grammatical factors which influence speakers' preference for one or another allomorph in the given context. If such factors are indeed non-existent, we might be dealing with an instance of true optionality. At the same time, there might be extragrammatical factors governing the choice of whether the ι boundary is deleted or not. To that effect, the only tangible difference between the chosen forms, according to the consultants, is in register. More specifically, speakers prefer the plural form in literary register while in the colloquial register, both forms can be used. The idea is that given the prescriptive bias, the use of the literary form is still allowed in the colloquial speech, while using a colloquial form in high register contexts is not possible. One piece of evidence supporting this claim is that in written texts, in plural environments, only the form *-oaqqii* is found. The idea is that two different mechanics of allomorph selection are used in the two registers. More precisely, in the formal register, the plural allomorph is universally selected in plural environments, while in the colloquial register, the allomorph selection in plural contexts is strictly sensitive to locality: the plural allomorph is selected with lower (more local) adjectives the number-neutral allomorph is selected with higher (less local) adjectives. This assumption presupposes that the two registers involve different allomorph selection rules: in the high register, the ι boundary is not deleted, while in the colloquial register, the iota boundary *is* deleted and the plural form is selected in plural contexts within ι, while the *zwamig* is the elsewhere form. This can be represented as follows.[14]

[14] Finally, one can assume that the same adjective – in our case, the adjective *small* – may merge at different heights in the nominal structure in non-idiomatic contexts. Since allomorph selection is known to be sensitive to locality, the plural allomorph can be hypothesized to be selected with lower attachment and the number neutral one – with higher

(26) **Colloquial (non-high) register: delete the ι boundary**

√SMALL ↔ kegii / [_ N[-SG]]$_{\omega}$

√SMALL ↔ zwamig

(27) **High register: don't delete the ι boundary**

√SMALL ↔ kegii / [_ N[-SG]]$_{\omega}$

√SMALL ↔ zwamig

3.3. √BIG

Now let's turn to the plural marking in *-oaqqa/-oaqqii*. The idea is that plural marking in this adjective involves two different and independent mechanics: concord and number-sensitive word-external contextual allomorph selection. None of these mechanics is unique to this adjective: to recall, prefixal number concord in this adjective is obligatory, similar to any other prefix-taking nominal modifier in the language. To that extent, there is no need to suggest any rule specific to one adjective (viz. *-oaqqa/-oaqqii*), contra Norris' proposal. As for the stem alternation, the same pattern of plural marking is found in *zwamig/kegii*. To that effect, both concord and number-sensitive allomorphy are found independently in the language and the number paradigm of *-oaqqa/-oaqqii* is only special in that it combines the two means of plural marking. This is illustrated below.

(28) Table 5

	Prefix number marking	Suppletive number marking
Prefix-taking adjectives (excl. *-oaqqa/-oaqqii*)	✔	✕
Non prefix-taking adjectives (excl. *zwamig/kegii*)	✕	✕
zwamig/kegii	✕	✔
-oaqqa/-oaqqii	✔	✔

I suggest that the rule governing the distribution of the two allomorphs of √BIG parallels that for *zwamig* vs. *kegii* mutatis mutandis for the root; concord plays no role for the allomorph selection. I remain agnostic on the relative timing of allomorph selection vs. concord. For explicitness, the allomorphic rule is provided below.

attachment. I have failed to find any independent evidence to support this assumption and this option thus remains a pure theoretical possibility.

(29) **Allomorph Selection Rule √BIG (ω – locality domain)**

√BIG ↔ *-oaqqii* / [_ N[-SG]]$_{\omega}$

√BIG ↔ *-oaqqa*

3.4. Against Different Root Analysis

Before turning to the discussion, it is important to consider – and ultimately refute – an alternative analysis of the patterns exhibited by the two studied Ingush adjectives. This analysis (Different Root Analysis, henceforth – DRA, see Borer (2014) for a discussion) would treat the forms *kegii* vs. *zwamig* and *–oaqqa*/*–oaqqii* as two different roots with no morphological relation to each other. This analysis may perhaps seem more plausible for *zwamig/kegii*, a clear case of full suppletion (with the two forms phonologically and diachronically unrelated), than for *–oaqqa/–oaqqii*. However, I will provide evidence against such an analysis for both pairs.

To start, one piece of evidence against such an analysis can be drawn from the data discussed earlier in the paper – namely, the fact that the same idiomatic meaning is shared by either form in the two pairs. To that effect, the idiomatic meaning pertaining to old age and/or high social status is found both with *-oaqqa* and *-oaqqii*. Similarly, the idiomatic meaning pertaining to young age is found both in *zwamig* and *kegii*. This is illustrated below.

(30) **Idiomatic meaning 'old', 'high social status'**

a. v-oaqqa sag
SG-big man
'a young person'

b. b-oaqqii nax
PL-big.PL people
'young people'

(31) **Idiomatic meaning 'young'**

a. zwamig sag
small man
'a young person'

b. kegii nax
small.PL people
'small people'

Next, for both adjectives, different forms can be used in ellipsis contexts. More specifically, the elided part may contain the mismatched form with respect to number, as the following examples show. This test is seen as a diagnostic for SRA since, under standard assumptions, the ellipsis requires root identity (see Harley (2015); Choi and Harley (2019)).

(32) **Mismatch in ellipsis with √SMALL**

a. c'agha cwa zwamig q'oalam b-oal-i je dukkxa ~~[kegii q'oalam-až]~~ ?
   at.home one small pen SG-be-Q CONJ many [small.PL pen-PL]
   'Is there one small pen in the house or many small pens?'

b. c'agha dukxa kegii q'oalam-až d-oaxk-i je cwa ~~[zwamig q'oalam]~~ ?
   at.home many small pen-PL PL-be.PL-Q CONJ one [small.SG pen]
   'Is there many small pens in the house or one?'

(33) **Mismatch in ellipsis with √BIG**

a. c'agha cwa b-oaqqa q'oalam b-oal-i je dukkxa ~~[d-oaqqii q'oalam-až]~~ ?
   at.home one SG-big pen SG-be-Q CONJ many [PL-big.PL pen-PL]
   'Is there one big pen in the house or many big pens?'

b. c'agha dukkxa d-oaqqii q'oalam-až d-oaxk-i je cwa ~~b-oaqqa q'oalam]~~ ?
   at.home many PL-big pen-PL PL-be.PL-Q CONJ one [SG-big pen]
   'Is there many big pens in the house or one?'

4. Suppletion: mediated or non-mediated by concord?

The allomorphy analysis of suppletion presented above raises the question of whether such allomorphy is directly sensitive to the number of the corresponding nominal (the head noun in the case of suppleting adjectives) or whether such allomorphy may be (or must be) mediated by an Agree (concord) relation.

Although on the theoretical level, this issue may be far from being resolved, several works provide empirical evidence that at least in some cases, number suppletion isn't plausibly modeled as mediated by Agree, or a similar morphosyntactic operation. For instance, as far as verbal suppletion is concerned, in certain languages, a given verb may cross-reference phi-features of one argument and show allomorphic sensitivity to the features of another. Such position is found for instance in Kolyma Yukaghir, among other languages, where the allomorphy tracks the person (1/2) of the recipient, while the agreement tracks the person of the direct object (3 person).

(34) **Kolyma Yukaghir**

met-in	er-če	n'ēr-ek	**kej-ŋile.**
I-DAT	[bad-ATTR]	clothing-PRED	give.1/2-OF:3SG

'They gave me bad clothing.' (Maslova 2002:353)

In addition to that, number suppletion may be found in languages without agreement (including Hiaki). Weisser (2019) attempts at providing of a list of features in which suppletion and agreement are demonstratively different. The available data on the phenomenon also shows that the locality of Agreement and feature-conditioned allomorphy might be different: while there are known cases of long-distance agreement there has been not described any cases of allomorphy sensitive to features of a DP in a different clause.[15]

The question of whether number suppletion is mediated by concord (or agreement) has been less prominently discussed for *adjectival* number suppletion (see however Adamson (2024)). As far as the Ingush data are concerned, we have seen that concord and number suppletion present two distinct phenomena, each with its distinct distributional signature. In our view, for Ingush, adjectival number suppletion is best analyzed as an unmediated operation, that is, as dissociated and independent from concord, which effectively accounts for the discrepancies found between the two operations.

The question remains however of whether Ingush adjectival number suppletion can still be modeled as a concord-mediated operation. An anonymous reviewer points out that concord is known to exhibit certain language specific idiosyncrasies, citing Russian concord with hybrid nouns, Arabic so called *deflected agreement* and optional concord in Miya as examples. The reviewer suggests that concord-mediated view can still be maintained if the feature exponence on the prefix (which marks cumulatively noun class and number) is treated as the result of one, obligatory, concord operation, and the non-prefixal number marking – as a reflex of another, optional, concord operation, which deals purely with number features.

First, it should be noted that prefixal concord in Ingush is standardly analyzed as involving a *cumulative* exponence of *two types of features* (viz. noun class and number) rather than exponing one feature (see e.g. Norris (2022:16), Desheriev and Desherieva (1999) for Ingush and Rudnev (2020) for other Nakh-Dagestanian languages). Note, for instance, that Class 6 nouns in Table 1 trigger different prefixes on

[15] See however Thornton (2019), Adamson (2024) who analyze number suppletion as Agree-mediated.

adjectives in singular and in plural, which shows that prefix marking can be sensitive to strictly number-related oppositions.

Relatedly, and perhaps, more importantly, those few modifiers that exhibit suffixal plural marking (e.g. *jer jerаž* 'this – these', some numerals e.g. *ši – šiaž* `two') do not exhibit optionality effects, which suggests that adnominal exponence of plurality, outside of stem suppletion, is uniformly obligatory. To that effect, adnominal suffixal plurality has the same distribution and properties as prefixal plurality (it strictly tracks the morphosytactic number of the head noun).

(35) **Obligatory number concord realized on suffixes**

a. jer q'oalam
this pen
'this pen'

b. jer-až/*jer q'oalam-až
this-PL/*this pen-PL
'these pens'

What that means is that if number suppletion in Ingush is indeed mediated by concord, a single set of features (viz. number) from the same controller (nominal head) manifests itself on two types of targets (prefix and suffix vs. stem alternations) in two different ways, showing obligatoriness in one case and optionality + locality-sensitivity in the other case. Although I do not know whether one can dismiss this theoretical possibility entirely in a strict sense, I can only point out that such position would present a highly non-typical picture with no clear precedent in the typological literature.

5. Conclusion

In this paper, I have considered in detail the patterns of double plural marking on the Ingush adjective -*oaqqa/-oaqqii* 'big'. The available data indicate that the two types of plurality have a different distribution and are most probably of a different nature. The non-prefixal plural marking on the adjective is analyzed in terms of allomorphy and is shown to have a clear analogue in the paradigm of another size adjective, *zwamig/kegii* 'small'.

The emerging picture points to the conclusion that the concord and allomorphy are two different operations and the latter cannot be (at least easily) reduced to the former. In that way, the data from Ingush can be directly compared to the analoguous body of evidence from the verbal domain where discrepancies between number suppletion and predicative agreement have equally been reported.

This conclusion doesn't imply however that every case of adnominal number-sensitive suppletion should be viewed as non-mediated allomorphy. One interesting case in this regard involves patterns of suppletion in Ingush demonstratives. To that effect, the demonstrative *yz/yž* ‘this/these’ involves a suppletive number paradigm, see below.[16]

(36) **Suppletive paradigm of proximal demonstratives *yz - yž***

a.	yz/*yž	q'oalam
	DEM.SG/*DEM.PL	pen
	‘this pen’	
b.	yž/*yz	q'oalam-až
	DEM.PL/*DEM.SG	pen-PL
	‘these pens’	

This pattern remarkably deviates from the one observed with the two discussed size adjectives. First, there is no optionality observed in plural contexts, unlike with -*oaqqa*. This is even more surprising, given that, if the suppletion in Ingush adjectives is indeed sensitive to locality, as suggested above – and given that demonstratives are generated very high on the nominal spine, higher than lexical adjectives. To wit, high modifiers are expected to exhibit more optionality with respect to allomorphy, not less. One plausible way to analyze these facts is to treat them as allomorphy which is mediated by concord.

The question of whether other instances of number suppletion can be seen as non-mediated allomorphy is left for future research.

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[16] At least historically, however, the two forms are related to each other, with the final consonantal segment -*ž* of the plural pronoun undoubtably deriving from the regular suffix -*až*, and the whole paradigm should be considered partial, rather than full, suppletion.

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